Check for updates

Hypertension

ORIGINAL ARTICLE

Nocturnal Hypertension and Prognosis in Patients of Very Advanced Age

Takeshi Fujiwara, Satoshi Hoshide, Kazuomi Kario.

BACKGROUND: Nocturnal blood pressure (BP) is a better predictor of health outcomes than office or daytime BP. However, the clinical significance of nocturnal hypertension in patients of very advanced age remains unexplored. We aimed to assess the association between nocturnal hypertension and composite cardiovascular outcomes in this population.

METHODS: This was a prospective observational study including Japanese elderly outpatients aged ≥80 years. All patients underwent 24-hour ambulatory BP monitoring at baseline. Nocturnal hypertension was defined as nocturnal systolic BP ≥120 mm Hg and diastolic BP≥70 mm Hg. Daytime hypertension was defined as daytime systolic BP ≥135 mm Hg and diastolic BP ≥85 mm Hg. The association between those BP phenotypes and composite cardiovascular outcomes (including fatal and nonfatal cardiovascular disease and all-cause mortality) was examined using Cox regression analysis.

RESULTS: A total of 485 patients were followed for a median of 3.9 years (1734 person-years), during which 72 (14.8%) composite cardiovascular outcomes occurred. The median age (interquartile range) was 82 (81–85) years; 44.7% were male; and 89.3% took antihypertensive medications. Nocturnal hypertension and daytime hypertension were present in 54.2% and 33.6% of patients, respectively. Relative to nocturnal normotension (nocturnal systolic BP <120 mm Hg and diastolic BP <70 mm Hg), nocturnal hypertension was associated with an increased risk of composite cardiovascular outcomes, even after adjustment for daytime BP values (adjusted hazard ratio, 2.15 [95% CI, 1.18–3.93]). Daytime hypertension showed no such association.

CONCLUSIONS: Screening for nocturnal hypertension identifies a high-risk group for composite cardiovascular outcomes among patients of very advanced age. ***(Hypertension.* 2026;83:00–00. DOI: 10.1161/HYPERTENSIONAHA.125.25199.)** • **Supplement Material**.

Key Words: aging ■ blood pressure ■ cardiovascular diseases ■ hypertension ■ phenotype

Hypertension FIRST PROOF ONLY

Downloaded from http://ahajournals.org by on February 11, 2026

Hypertension is the most important modifiable risk factor for cardiovascular disease (CVD).[1,2] Most hypertension guidelines have recommended that the diagnosis and management of hypertension should be made based on patients' out-of-office blood pressure (BP) levels.[3–6] A major advantage of ambulatory BP monitoring (ABPM) over conventional office BP measurement is its ability to evaluate 24-hour BP levels and circadian BP rhythm. Among BP parameters assessed by ABPM, nocturnal BP is the strongest predictor of CVD outcomes,[7,8] and nocturnal hypertension is regarded as an important predictive factor for cardiovascular outcomes in patients with hypertension.[9] Therefore, the identification of nocturnal hypertension in daily clinical practice is crucial for improving CVD risk stratification.

Previous randomized controlled trials have demonstrated that intensive BP reduction can improve cardiovascular outcomes in elderly patients with hypertension.[10–12] However, the benefit of intensive BP control in patients with hypertension aged ≥80 years remains controversial.[13,14] Moreover, the association between nocturnal hypertension and cardiovascular outcomes has not been evaluated in this age group, and the relative prognostic impact of nocturnal versus daytime hypertension also remains unclear in patients of very advanced age.

Correspondence to: Kazuomi Kario, Division of Cardiovascular Medicine, Department of Medicine, Jichi Medical University School of Medicine, 3311-1, Yakushiji, Shimotsuke, Tochigi 329-0498, Japan. Email kkario@jichi.ac.jp
Supplemental Material is available at https://www.ahajournals.org/doi/suppl/10.1161/HYPERTENSIONAHA.125.25199.
For Sources of Funding and Disclosures, see page XXX.
© 2026 American Heart Association, Inc.
Hypertension is available at www.ahajournals.org/journal/hyp

NOVELTY AND RELEVANCE

What Is New?

We demonstrated that nocturnal hypertension is associated with increased risk for composite cardiovascular outcome, independently of daytime blood pressure levels, in patients of very advanced age (≥80 years).

What is Relevant?

These findings highlight the importance of 24-hour ambulatory blood pressure monitoring in the very elderly population to detect nocturnal hypertension, which may otherwise be missed by office or daytime measurements.

Clinical/Pathophysiological Implications?

Nocturnal hypertension was associated with a higher risk of composite cardiovascular outcomes in individuals aged ≥80 years. Assessment of nocturnal blood pressure levels may improve cardiovascular risk stratification in this population and inform clinical decision-making.

Nonstandard Abbreviations and Acronyms

ABP-International	Ambulatory Blood Pressure in Referred Hypertensive Subjects: an International Database Study
ABPM	ambulatory blood pressure monitoring
BP	blood pressure
CVD	cardiovascular disease
DBP	diastolic blood pressure
HR	hazard ratio
SBP	systolic blood pressure
SEARCH	Search Longevity in Very Elderly with Ambulatory Pressure in Tochigi

Using data from the Japanese-based study known as the SEARCH study (Search Longevity in Very Elderly with Ambulatory Pressure in Tochigi), a prospective observational study of elderly patients (≥80 years old), we aimed to (1) examine the association between nocturnal hypertension and the risk of composite cardiovascular outcome in patients of very advanced age; and (2) to compare the risk associated with nocturnal hypertension and daytime hypertension in the same study population. We hypothesized that nocturnal hypertension would be associated with increased risk for composite cardiovascular outcomes compared with daytime hypertension in patients of very advanced age (≥80 years).

METHODS

Ethics Committee Approval and Informed Consent

The institutional review board of Jichi Medical University School of Medicine approved the methods, and all patients provided written informed consent to participate and to have their data published.

In accordance with the Transparency and Openness Promotion Guidelines, we note that this study was not preregistered. However, the research objectives, study design, and analysis plan were determined before data analysis and were not modified thereafter.

The data, analytic methods, and study materials that support the findings of the current study are available from the corresponding author on reasonable request.

Study Design

Details of the SEARCH study rationale, design, and procedures have been published previously.[15] Briefly, 525 elderly outpatients were recruited between September 2008 and December 2013 and followed up through April 2017 by 26 doctors at 21 institutions (including 2 specialized university hospitals). Details of the study design and methods are described in the Supplemental Material of the present study.

The 3 inclusion criteria were (1) age ≥80 years old; (2) ability to attend a clinic or hospital without difficulty in walking; and (3) living without assistance. The exclusion criteria were (1) cardiovascular or cerebrovascular events occurring within the past 6 months, excluding transient ischemic attack; (2) current dialysis; (3) malignant disease at baseline; and (4) inability to provide informed consent due to severe cognitive dysfunction or dementia.

BP and Other Measurements

Details of the methods are described in the Supplemental Material. Two office BP readings were taken with a 1–2 min interval, and the average of the measurements was used as the office BP value.[16] Noninvasive 24-hour ABPM was carried out at the baseline with a validated automatic device (TM-2425 or TM-2431; A&D Co, Tokyo) that recorded the patient's BP at 30-minute intervals throughout the 24-hour day using an oscillometric method. A 24-hour ABPM recording was considered acceptable if at least 70% of the measurements were valid.[17] Nocturnal BP was defined as the average BP of the values recorded during the period beginning when patients went to bed and ending when they woke in the morning, and daytime

Downloaded from http://ahajournals.org by on February 11, 2026

BP was defined as the average BP value for the rest of the day. The nighttime sleep period was determined by self-report using a sleep diary.

Fasting blood and spot urine samples were collected in the morning at baseline. All samples were sent to a single laboratory (SRL, Tokyo) within 24 hours of collection. Diabetes was defined as self-reported history of a physician's diagnosis, diabetes medication use, fasting blood glucose level ≥126 mg/dL, or nonfasting glucose level ≥200 mg/dL. Chronic kidney disease was defined as the presence of proteinuria or a value of <60 mL/min/1.73m^2 for the estimated glomerular filtration rate.[18] History of heart failure was defined as preexisting congestive heart failure. History of CVD was defined as preexisting angina pectoris, myocardial infarction, or stroke.

Definition of BP Phenotype

We set the cut off thresholds of office and ambulatory BP based on hypertension guidelines.[4,5] Nocturnal hypertension was defined as hypertensive nocturnal BP (nocturnal systolic BP [SBP] ≥120 mm Hg and nocturnal diastolic BP [DBP] ≥70 mm Hg); nocturnal normotension as normotensive nocturnal BP (nocturnal SBP <120 mm Hg and nocturnal DBP <70 mm Hg); daytime hypertension as hypertensive daytime BP (daytime SBP ≥135 mm Hg and daytime DBP ≥85 mm Hg); and daytime normotension as normotensive daytime BP (daytime SBP <135 mm Hg and daytime DBP <85 mm Hg). In addition, we subclassified the study patients into 4 BP phenotypes: isolated nocturnal hypertension, defined as hypertensive nocturnal BP and normotensive daytime BP; isolated daytime hypertension, defined as hypertensive daytime BP and normotensive nocturnal BP; sustained hypertension, defined as hypertensive nocturnal and daytime BPs; and controlled BP, defined as normotensive nocturnal and daytime BPs.

We previously reported that a riser BP pattern was significantly associated with composite cardiovascular outcomes, independently of the mean nocturnal BP level, in the same study population.[19] Patients with a reduced nocturnal BP dip typically exhibit higher nocturnal BP levels,[20] indicating a close relationship between these 2 parameters. Therefore, in this study, we aimed to evaluate the risk of composite cardiovascular outcomes according to the presence of nocturnal hypertension and a riser BP pattern. The nocturnal SBP fall (%) was calculated as 100×(1− nocturnal SBP/daytime SBP). We defined a riser BP pattern as a nocturnal SBP reduction rate of <0%.[21]

Ascertainment of Outcomes

We defined a composite cardiovascular outcome comprising coronary heart disease (including fatal and nonfatal coronary artery disease, and sudden death within 24 hours of symptom onset), stroke (fatal and nonfatal), congestive heart failure (fatal and nonfatal), aortic dissection (fatal and nonfatal), and all-cause mortality. In cases with multiple events, only the first event was included in the analysis.

Each patient's vital status was ascertained through April 2017. Cardiovascular outcomes were ascertained through ongoing reports from general physicians at each participating institution. Patients were followed until the first composite cardiovascular outcome or otherwise censored at the last study visit.

Statistical Analyses

Descriptive statistics are presented as the mean and SD or median with interquartile range where appropriate for continuous variables, and as frequencies and percentages for categorical variables. Demographic variables and clinical characteristics were compared between the nocturnal or daytime BP phenotypes (nocturnal normotension versus nocturnal hypertension, and daytime normotension versus daytime hypertension) using the Wilcoxon rank-sum test for continuous variables and the χ^2 test for categorical variables.

Cumulative incidence estimates were used to compare raw (unadjusted) and adjusted composite cardiovascular outcomes between BP phenotypes using Gray's test and Fine–Gray regression, respectively, to account for the competing risk of all-cause death. Adjusted factors included age, gender, body mass index, smoking status, prevalence of diabetes, prevalence of dyslipidemia, prevalence of chronic kidney disease, history of CVD, number of antihypertensive medications, office SBP, and either (1) daytime SBP when we used nocturnal BP to define BP phenotype, or 2) nocturnal SBP when we used daytime BP to define BP phenotype.

Cox proportional hazards models were used to calculate hazard ratios (HRs) and 95% CIs for the likelihood of experiencing a composite cardiovascular outcome among BP phenotypes. Schoenfeld residuals were used to confirm the proportional hazards assumption. HRs were calculated in an unadjusted model (model 1). Model 2 included adjustment for the covariates mentioned above, and model 3 included model 2 components and daytime SBP or nocturnal SBP. These covariates were selected a priori because they have been reported to show correlations with both BP[22] and CVD risk[23] and could potentially confound the association between BP and CVD risk.

In the analysis of the association between the outcomes of this study and continuous variables, we first assessed the linearity of the associations between nocturnal SBP or DBP levels and the risk of composite cardiovascular outcomes in Cox regression. We then calculated HRs by tertiles of each BP variable. To further explore the shape of the relationship between nocturnal BP levels and composite cardiovascular events risk, we applied restricted cubic spline regression with 4 knots at prespecified percentiles (25th, 50th, 75th, and 95th) of the nocturnal BP distributions. The resulting curves illustrate the HRs and 95% CIs calculated relative to the chosen reference values, which correspond to the mean nocturnal SBP and DBP values in the study population.

In a sensitivity analysis, we examined the association between nocturnal hypertension and a modified composite cardiovascular outcome that excluded congestive heart failure. This approach allowed us to assess whether nocturnal hypertension is truly associated with overall cardiovascular risk or merely reflects a fluid overload phenotype commonly seen in elderly individuals. In addition, we evaluated the association of presleep BP with a composite cardiovascular outcome, which helped to distinguish whether the risk is driven by nocturnal SBP during the sleeping period itself or by presleep BP levels just before bedtime. Presleep BP was defined as BP measured just before going to bed, and presleep hypertension was defined as presleep SBP ≥120 mm Hg and presleep DBP ≥70 mm Hg.

All statistical analyses were performed with R software, ver. 4.3.2 (The R Foundation for Statistical Computing, Vienna,

Downloaded from http://ahajournals.org by on February 11, 2026

Hypertension FIRST PROOF ONLY

Austria). Two-sided $P<0.05$ was defined as statistically significant.

RESULTS

Of the 525 patients for whom entry data were obtained, we excluded 5 patients who did not have ABPM data, 3 patients who were lost to follow-up, and 32 patients who had ≥1 missing covariate, leaving a sample of 485 patients for analysis. Table 1 shows baseline characteristics of the SEARCH study participants included in the present analysis. The median age (interquartile range) of the 485 participants was 82 (81–85) years, 44.7% were male, and 89.3% were using antihypertensive medications. The total number of ABPM recordings was 40.5±6.5, including 15.0±4.3 during the nighttime period. For each ABPM recording, the proportion of valid readings exceeded 70% of the total 48 measurements, thereby fulfilling the quality criteria.[17] Although the difference in nocturnal SBP levels between nocturnal and daytime hypertension was clinically small, patients with daytime hypertension had daytime SBP levels ≈10

Table 1. Baseline Characteristics of the Study Patients

Variable	Overall (n=485)	Controlled BP (n=209)	Isolated daytime hypertension (n=13)	Isolated nocturnal hypertension (n=113)	Sustained hypertension (n=150)	*P* value
Descriptive variables						
Age, y	82 (81–85)	82 (80–85)	81 (81–83)	83 (81–85)	82 (81–85)	0.653
Male, n (%)	217 (44.7)	77 (36.8)	3 (23.1)	62 (54.9)	75 (50.0)	0.003
Body mass index, kg/m^2	23.1 (21.2–25.2)	23.1 (21.3–25.0)	22.1 (21.5–24.4)	22.2 (20.0–24.5)	23.6 (21.8–25.8)	0.013
Current smoker, n (%)	27 (5.6)	12 (5.7)	0 (0.0)	4 (3.5)	11 (7.3)	0.466
Daily drinker, n (%)	124 (25.6)	57 (27.3)	3 (23.1)	28 (24.8)	36 (24.2)	0.907
Diabetes, n (%)	124 (25.6)	49 (23.4)	5 (38.5)	26 (23.0)	44 (29.3)	0.371
Chronic kidney disease, n (%)	167 (34.4)	61 (29.2)	3 (23.1)	44 (38.9)	59 (39.3)	0.116
History of heart failure, n (%)	97 (20.1)	50 (24.2)	2 (15.4)	22 (19.5)	23 (15.3)	0.217
History of CVD, n (%)	117 (24.1)	45 (21.5)	4 (30.8)	25 (22.1)	43 (28.7)	0.389
Fasting glucose, mg/dL	100 (91–115)	101 (92–114)	103 (96–122)	99 (91–114)	100 (90–119)	0.553
Total cholesterol, mg/dL	188 (170–212)	189 (170–212)	195 (184–212)	185 (169–202)	190 (169–215)	0.304
HDL cholesterol, mg/dL	55 (46–65)	56 (48–66)	65 (51–68)	53 (44–63)	55 (45–65)	0.072
Antihypertensive medication use, n (%)	433 (89.3)	188 (90.0)	13 (100.0)	96 (85.0)	136 (90.7)	0.244
Calcium-channel blockers, n (%)	315 (64.9)	144 (68.9)	9 (69.2)	63 (55.8)	99 (66.0)	0.121
ACE inhibitors, n (%)	114 (23.5)	39 (18.7)	3 (23.1)	23 (20.4)	49 (32.7)	0.016
Angiotensin receptor blockers, n (%)	236 (48.7)	116 (55.5)	7 (53.8)	46 (40.7)	67 (44.7)	0.049
Diuretics, n (%)	132 (27.2)	70 (33.5)	3 (23.1)	28 (24.8)	31 (20.7)	0.049
Beta-blockers, n (%)	89 (18.4)	40 (19.1)	6 (46.2)	17 (15.0)	26 (17.3)	0.052
Alpha-blockers, n (%)	43 (8.9)	18 (8.6)	0 (0.0)	6 (5.3)	19 (12.7)	0.125
No. of antihypertensive medications	3 (2–4)	3 (2–4)	3 (2–4)	3 (2–3)	3 (2–4)	0.013
BP measures, mmHg						
Office SBP	139.0 (126.0–155.0)	135.0 (122.0–148.0)	149.0 (146.0–163.0)	138.0 (127.0–151.0)	149.0 (134.3–163.8)	<0.001
Office DBP	73.0 (64.0–83.0)	70.0 (62.0–81.0)	85.0 (72.0–88.0)	74.0 (68.0–82.0)	76.0 (67.0–84.0)	0.004
24-h SBP	129.6 (120.2–138.4)	120.0 (113.9–125.9)	136.9 (135.5–137.0)	129.2 (125.5–132.5)	144.4 (139.0–150.9)	<0.001
24-h DBP	70.3 (65.6–75.6)	66.4 (62.7–69.3)	72.9 (70.3–76.1)	71.7 (67.7–76.3)	75.9 (72.3–80.4)	<0.001
Daytime SBP	134.0 (124.9–144.6)	126.5 (119.1–133.7)	149.5 (143.9–150.9)	130.4 (124.0–135.5)	148.4 (143.6–157.6)	<0.001
Daytime DBP	73.2 (67.8–78.5)	69.0 (65.8–73.5)	80.0 (76.9–82.1)	72.5 (67.8–78.1)	78.6 (74.9–83.1)	<0.001
Nighttime SBP	120.2 (110.7–130.9)	109.1 (101.9–114.6)	112.7 (112.2–115.4)	125.0 (121.1–129.5)	135.1 (128.8–147.1)	<0.001
Nighttime DBP	65.3 (59.9–71.9)	60.3 (56.6–63.8)	61.2 (56.4–67.2)	70.7 (65.0–73.6)	71.3 (67.0–75.5)	<0.001

Data are median (interquartile range) or n (%). ACE indicates angiotensin converting enzyme; CVD, cardiovascular disease; DBP, diastolic blood pressure; HDL, high-density lipoprotein; SBP, systolic blood pressure. Chronic kidney disease was defined as estimated glomerular filtration rate <60 mL/min/1.73m^2 or albuminuria, which was defined as urine albumin-to-creatinine ratio ≥30 mg/g·Cr. History of CVD included preexisting angina pectoris, myocardial infarction, or stroke.

Hypertension FIRST PROOF ONLY

Downloaded from http://ahajournals.org by on February 11, 2026

mm Hg higher than those with nocturnal hypertension (Table S1). The demographic variables and clinical characteristics for the BP phenotypes defined by the presence or absence of nocturnal hypertension and a riser BP pattern are shown in Table S2.

During a median (interquartile range) follow-up of 3.9 (2.3–4.3) years (1734 person-years), 72 composite cardiovascular outcomes (41.5 per 1000 person-years) occurred. The 72 composite cardiovascular outcomes consisted of 7 coronary heart disease, 19 stroke, 14 congestive heart failure, and 1 aortic dissection event(s), and 31 all-cause deaths (age-related: n=8; cancer: n=12; pneumonia: n=7; unknown: n=4). The cumulative incidence of composite cardiovascular outcomes was significantly higher in patients with nocturnal hypertension compared with those with nocturnal normotension: 52.0 per 1000 person-years (95% CI, 39.3–68.7) versus 28.9 per 1000 person-years (95% CI, 19.3–43.4; Figure 1A). This association was also observed even after adjustment for covariates, including daytime SBP levels (Figure 1C). In contrast, no significant difference was observed between patients with daytime hypertension and those with daytime normotension: 46.2 per 1000 person-years (95% CI, 31.8–67.3) versus 39.0 per 1000 person-years (95% CI, 29.2–52.2; Figure 1B and 1D).

After adjustment for covariates, results from the Cox proportional hazards models indicated that nocturnal hypertension was associated with a significantly higher risk of composite cardiovascular outcomes compared with nocturnal normotension (adjusted HR, 2.04 [95% CI, 1.20–3.49]). This association remained significant after additional adjustment for daytime SBP levels (adjusted HR, 2.15 [95% CI, 1.18–3.93]; Table 2). In contrast, no significant association was observed between daytime hypertension and daytime normotension (Table 2).

When all BP variables were analyzed as continuous variables, nocturnal SBP tended to be associated with an increased risk of composite cardiovascular outcomes (Table S3). In the restricted cubic spline analysis, both

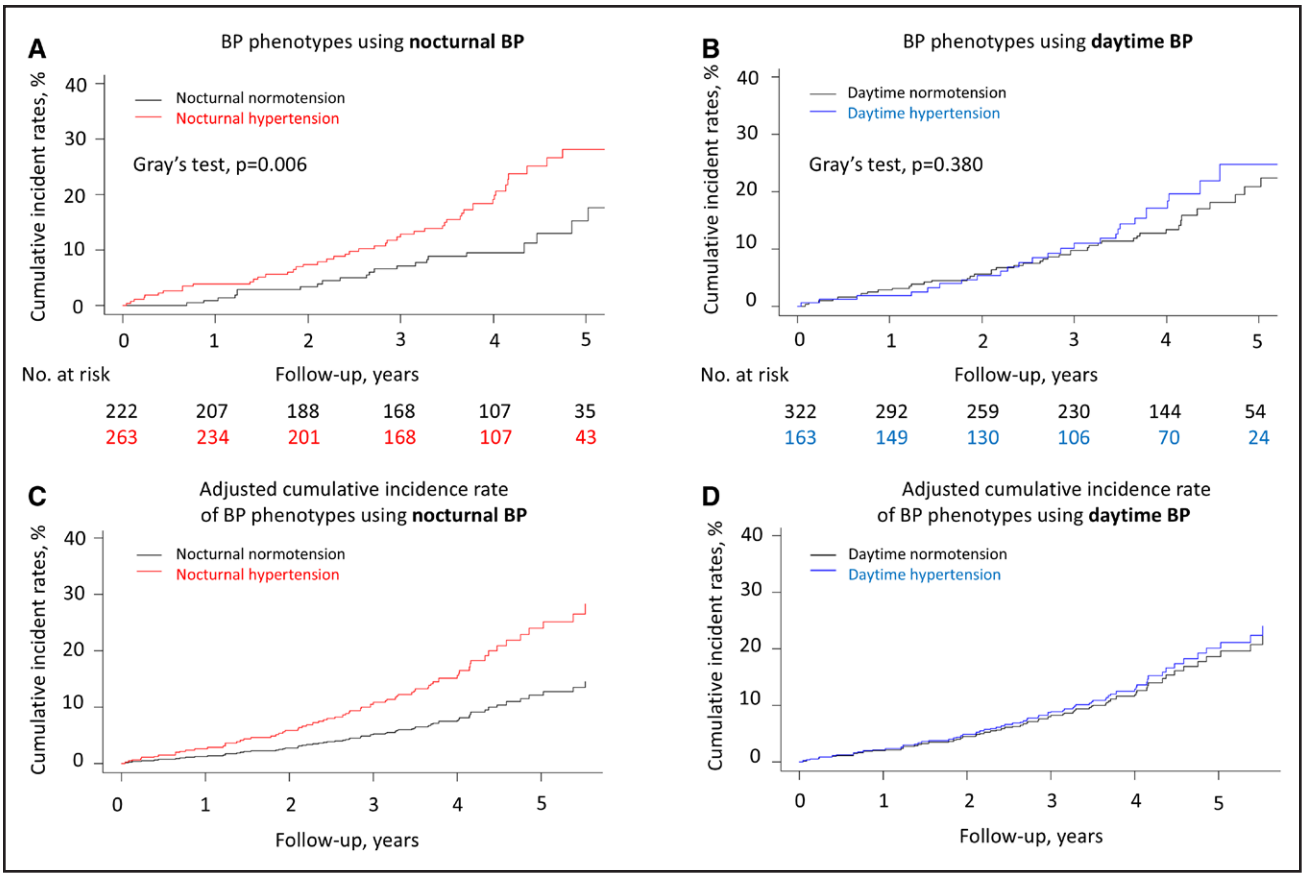

Figure 1. The blood pressure (BP) phenotypes of nocturnal or daytime hypertension and composite cardiovascular outcomes. Kaplan-Meier curves of the cumulative incidence of composite cardiovascular outcomes by each BP phenotype are shown. **A**, Unadjusted cumulative incidence rate in BP phenotypes defined using nocturnal BP. **B**, Unadjusted cumulative incidence rate in BP phenotypes defined using daytime home BP. **C**, Adjusted cumulative incidence rate in BP phenotypes defined using nocturnal BP. **D**, Adjusted cumulative incidence rate in BP phenotypes defined using daytime home BP. Adjusted factors included age, gender, body mass index, smoking status, prevalence of diabetes, prevalence of dyslipidemia, prevalence of chronic kidney disease, history of cardiovascular disease, number of antihypertensive medications, office systolic BP (SBP), and (**C**) daytime SBP or (**D**) nocturnal SBP. The numbers at risk are unavailable for adjusted survival curves, as the curves are adjusted proportionally with the confounding variables. Including the original numbers at risk would inaccurately reflect the curves because the scaling is applied to survival probabilities.

Downloaded from http://ahajournals.org by on February 11, 2026

Table 2. Hazard Ratio for Composite Cardiovascular Outcome by Nocturnal Hypertension and Daytime Hypertension

	Nocturnal hypertension		Daytime hypertension	
	No (n=222)	Yes (n=263)	No (n=322)	Yes (n=163)
No. of events	23	49	45	27
Rate (95% CI)*	28.9 (19.3–43.4)	52.0 (39.3–68.7)	39.0 (29.2–52.2)	46.2 (31.8–67.3)
Model 1 (unadjusted)	1 [Reference]	1.98 (1.21–3.25)	1 [Reference]	1.24 (0.77–2.00)
Model 2 (adjusted)	1 [Reference]	2.04 (1.20–3.49)	1 [Reference]	1.45 (0.86–2.42)
Model 3	1 [Reference]	2.15 (1.18–3.93)	1 [Reference]	1.13 (0.57–2.22)

The adjusted HRs (95% CIs) associated with nocturnal hypertension and daytime hypertension are shown. Nocturnal hypertension was defined as nocturnal SBP of 120 mm Hg or higher or DBP of 70 mm Hg or higher. Daytime hypertension was defined as daytime SBP of 135 mm Hg or higher or DBP of 85 mm Hg or higher. Adjusted factors for Model 2 included demographic variables (age and sex), clinical characteristics (body mass index, smoking status, prevalence of diabetes, prevalence of dyslipidemia, prevalence of chronic kidney disease, history of CVD, and number of antihypertensive medications), and office SBP. Adjustment factors for Model 3 included the Model 2 components and daytime SBP or nocturnal SBP. Composite outcome included coronary heart disease (fatal and nonfatal coronary artery disease and sudden death within 24 hours of the abrupt onset of symptoms), stroke (fatal and nonfatal), congestive heart failure (fatal and nonfatal), aortic dissection (fatal and nonfatal), and all-cause mortality. History of CVD included pre-existing angina pectoris, myocardial infarction, or stroke. CI indicates CI; CVD, cardiovascular disease; DBP, diastolic blood pressure; HR, hazard ratio; SBP, systolic blood pressure.

*The rate is per 1000 person-years.

nocturnal SBP and DBP exhibited nonlinear associations with cardiovascular risk. The association for nocturnal SBP showed a generally increasing but nonlinear pattern, whereas nocturnal DBP demonstrated a U-shaped relationship (Figure 2). When SBP levels were categorized into tertiles, the highest tertile of nocturnal SBP tended to be associated with an increased risk, while neither daytime SBP nor office SBP showed significant associations (Table 3). Similarly, when DBP levels were categorized into tertiles, none of the DBP variables were significantly associated with composite cardiovascular outcomes (Table S4). The cumulative incidence of composite cardiovascular outcomes across tertiles of SBP and DBP is shown in Figure S1.

When study patients were classified based on the presence of nocturnal and daytime hypertension, both isolated nocturnal hypertension and sustained hypertension were associated with a higher risk of composite cardiovascular outcomes compared with those with controlled BP: isolated nocturnal hypertension, adjusted HR, 2.31 [95% CI, 1.24–4.30]; sustained hypertension, adjusted HR, 2.10 [95% CI, 1.12–3.92] (Figure 3).

Furthermore, when patients were categorized according to the presence or absence of nocturnal hypertension and a riser BP pattern, those with nocturnal hypertension, regardless of the presence of a riser BP pattern, had an increased risk of composite cardiovascular outcomes compared with those without nocturnal hypertension and a riser BP pattern: nocturnal hypertension without a riser BP pattern, adjusted HR, 1.89 (95% CI, 1.06–3.34); nocturnal hypertension with a riser BP pattern, adjusted HR, 2.75 (95% CI, 1.35–5.60; Figure S2).

In the sensitivity analysis excluding congestive heart failure from the composite outcome, the association between nocturnal hypertension and cardiovascular risk remained essentially unchanged (Table S5). The mean presleep BP levels were 127.6±24.9 mm Hg of SBP and 69.5±14.6 mm Hg of DBP. Presleep BP levels were not associated with the increased risk of composite cardiovascular outcomes (Table S6), and hypertension defined by presleep BP was also related to the incidence of composite cardiovascular outcomes (Table S7).

Hypertension FIRST PROOF ONLY

DISCUSSION

Main Study Findings

In this prospective, practice-based observational study of 485 elderly Japanese outpatients of ≥80 years of age, we found that nocturnal hypertension was associated with increased risk for composite cardiovascular outcome, independently of daytime BP levels. Notably, despite having ≈10 mm Hg lower daytime SBP and similar nocturnal BP levels, patients with nocturnal hypertension (defined as nocturnal SBP ≥120 mm Hg and DBP ≥70 mm Hg) had a higher cardiovascular risk than those with daytime hypertension (daytime SBP ≥135 mm Hg and DBP ≥85 mm Hg), which was not associated with adverse outcomes, as shown Fig. 1 and Table 2. Furthermore, in a separate analysis based on combined BP phenotype (Figure 3), the magnitude of risk in patients with isolated nocturnal hypertension (hypertensive nocturnal BP and normotensive daytime BP) was comparable to that observed in patients with sustained hypertension (hypertensive nocturnal and daytime BP). To our knowledge, this is the first study to demonstrate the prognostic significance of nocturnal hypertension in patients of very advanced age (≥80 years) and to directly compare the risk of composite cardiovascular outcome between nocturnal and daytime hypertension within the same study population.

Downloaded from http://ahajournals.org by on February 11, 2026

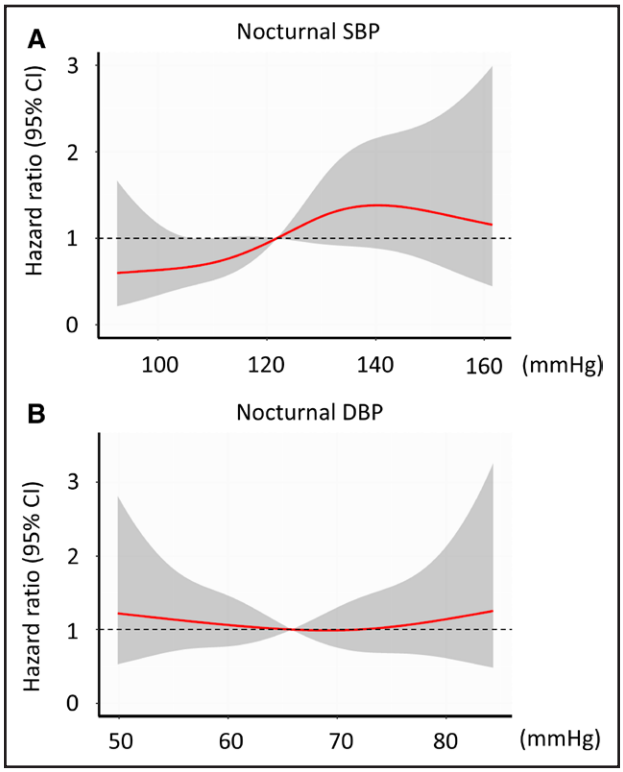

Figure 2. Association between nocturnal blood pressure (BP) levels and the risk of composite cardiovascular outcomes. The depicted model was fitted using a restricted cubic spline with 4 knots to model nocturnal BP levels, with adjustment for demographic variables (age and sex), clinical characteristics (body mass index, smoking status, prevalence of diabetes, prevalence of dyslipidemia, prevalence of chronic kidney disease, history of cardiovascular disease, and number of antihypertensive medications), and office systolic blood pressure (SBP). The curves show the hazard ratios compared with the chosen reference values corresponding to the mean nocturnal SBP and diastolic blood pressure (DBP) levels in the study population. The gray shaded areas around the curves indicate the 95% CIs. **A**, Association between nocturnal SBP and composite cardiovascular outcomes modeled by a restricted cubic spline, with a reference value of 121.8 mm Hg. **B**, Association between nocturnal DBP and composite cardiovascular outcomes modeled by a restricted cubic spline, with a reference value of 65.8 mm Hg.

These findings highlight the importance of assessing nocturnal BP levels to improve cardiovascular risk stratification in this population.

Comparison With Previous Literature

In the SEARCH study, patients with nocturnal hypertension had ≈2-fold higher risk of composite cardiovascular outcome compared with those with nocturnal normotension, whereas daytime hypertension was not associated with cardiovascular risk. These findings indicate that nocturnal BP contributes more substantially to composite cardiovascular outcomes than daytime BP. The Spanish Ambulatory Blood Pressure Registry, a large population-based cohort study (n=59 124 and 59.4% received

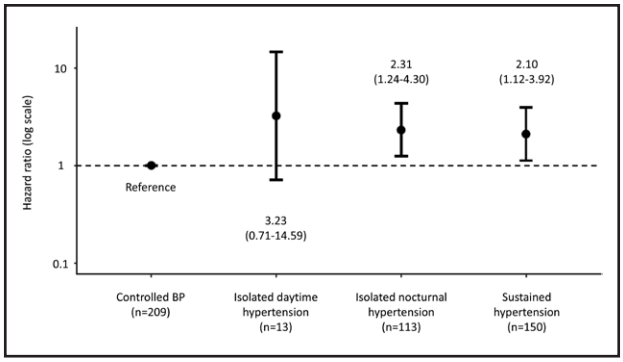

Figure 3. Adjusted hazard ratios for composite cardiovascular outcomes according to blood pressure (BP) phenotypes based on ambulatory BP monitoring. The adjusted hazard ratios (HRs) for composite cardiovascular outcomes in patients aged ≥80 years, categorized into 4 BP phenotypes based on ambulatory BP monitoring, are shown. The reference group is patients with controlled BP. Each dot represents the HR, and the vertical lines indicate the corresponding 95% CIs. Adjusted factors included age, gender, body mass index, smoking status, prevalence of diabetes, prevalence of dyslipidemia, prevalence of chronic kidney disease, history of cardiovascular disease, number of antihypertensive medications, and office SBP.

antihypertensive medication), showed that nighttime SBP was associated with an increased risk of all-cause and cardiovascular death even after adjustment for daytime SBP. Notably, the prognostic value of daytime SBP was lost after adjustment for nighttime SBP.[8] A similar pattern was reported in the Ambulatory Blood Pressure Collaboration in Patients Diagnosed With Hypertension, a multinational pooled cohort comprising 13 844 patients with hypertension from Europe, Brazil, and Japan. In that analysis, a 10 mm Hg increase in nighttime SBP was associated with a significantly higher risk of cardiovascular events, even after adjustment for daytime and office SBP. In contrast, the association between daytime SBP and cardiovascular outcomes was attenuated after simultaneous adjustment for nighttime SBP.[24] Our current findings extend these previous observations by demonstrating that nocturnal hypertension, but not daytime hypertension, was significantly associated with composite cardiovascular outcomes in patients of very advanced age (≥80 years), even after adjustment for daytime SBP levels. These results emphasize the clinical value of assessing nocturnal BP to improve risk stratification in very elderly patients with hypertension.

Our study demonstrated that isolated nocturnal hypertension was associated with an increased risk of composite cardiovascular outcomes. These findings are consistent with previous reports. An analysis of the International database of Ambulatory BP in relation to Cardiovascular Outcome database, a population-based study comprising 10 cohorts (mean age, 54.8 years, 47.0% female), showed that isolated nocturnal hypertension was associated with a higher risk of total mortality and all cardiovascular events compared with nocturnal

Hypertension FIRST PROOF ONLY

Downloaded from http://ahajournals.org by on February 11, 2026

Table 3. The Risk of Composite Cardiovascular Outcome by Tertiles of Nocturnal, Daytime, and Office SBP

	BP range, mm Hg	No. of events	No. of events per 1000 person-years (95% CI)*	HR (95% CI)	*P* value
Nocturnal SBP					
Tertile 1 (n=160)	<113.9	20	34.9 (22.6–53.9)	0.97 (0.51–1.81)	0.912
Tertile 2 (n=161)	≤113.9–<127.4	20	34.7 (22.4–53.6)	1 [Reference]	n/a
Tertile 3 (n=164)	127.4 ≤	32	54.5 (38.6–77.0)	1.69 (0.95–2.99)	0.073
Daytime SBP					
Tertile 1 (n=161)	<128.0	25	43.3 (29.4–64.0)	1.31 (0.71–2.42)	0.392
Tertile 2 (n=161)	≤128.0–<141.3	19	32.9 (21.1–51.4)	1 [Reference]	n/a
Tertile 3 (n=163)	≤141.3	28	47.9 (33.2–69.3)	1.70 (0.93–3.10)	0.083
Office SBP					
Tertile 1 (n=167)	<131.0	27	45.1 (31.0–65.7)	0.76 (0.43–1.34)	0.346
Tertile 2 (n=165)	≤131.0–<149.0	27	45.7 (31.4–66.4)	1 [Reference]	n/a
Tertile 3 (n=153)	≤149.0	18	32.8 (20.8–51.9)	0.69 (0.37–1.27)	0.227

Each reference was defined as the lowest no. of events per 1000 person-years. Adjusted factors included demographic variables (age and sex), clinical characteristics (body mass index, smoking status, prevalence of diabetes, prevalence of dyslipidemia, prevalence of chronic kidney disease, history of CVD, and no. of antihypertensive medications), and office SBP. BP indicates blood pressure; CVD, cardiovascular disease; HR, hazard ratio; and SBP, systolic blood pressure.

*No. of events (95% CI) per 1000 person-years.

normotension.[25] Similarly, in a cohort of Chinese patients with chronic kidney disease (n=1282, mean age, 44.0 years, 40.9% female), isolated nocturnal hypertension was associated with increased risk of cardiovascular events, even after adjustment for office BP, 24-hour BP, or daytime BP.[26] Our findings extend previous evidence by demonstrating that isolated nocturnal hypertension is also a significant cardiovascular risk factor in individuals aged ≥80 years. Given the high prevalence of nocturnal hypertension in the elderly, possibly due to age-related physiological changes such as impaired fluid regulation and circadian BP control, the clinical relevance of isolated nocturnal hypertension may be even greater in this population. In fact, the risk of composite cardiovascular outcomes in patients with isolated nocturnal hypertension was comparable to that in those with sustained hypertension. Further large-scale, prospective studies are warranted to investigate the prognostic value of isolated nocturnal hypertension in patients of very advanced age (≥80 years).

To differentiate the cardiovascular risk associated with nocturnal hypertension from that of abnormal circadian BP variability characterized by a riser BP pattern, we categorized patients into 4 groups based on the presence or absence of nocturnal hypertension and a riser BP pattern. Our findings showed that nocturnal hypertension was associated with an increased risk of composite cardiovascular outcomes, regardless of the presence of a riser BP pattern. These results are consistent with findings from the Spanish Ambulatory Blood Pressure Registry, which reported that nocturnal hypertension with normal dipping was associated with increased risks of all-cause and cardiovascular mortality, and that the highest risk was observed in individuals with both nocturnal hypertension and abnormal dipping patterns.[27] We previously demonstrated that a riser BP pattern was associated with a higher risk of composite cardiovascular outcomes, independent of nocturnal BP levels.[19] Taken together, these findings suggest that the pathophysiological mechanisms underlying nocturnal hypertension and a riser BP pattern may differ. Nocturnal hypertension may primarily reflect volume overload due to impaired renal sodium excretion or increased extracellular fluid volume,[28] while a riser BP pattern could indicate abnormal nocturnal sympathetic nervous system activation or baroreflex dysfunction.[29] These differences are particularly relevant in the very elderly, in whom age-related vascular stiffness,[30] neurohormonal and autonomic dysregulation,[31] renal sodium handling,[32] and sleep disturbances may modify the expression and impact of these BP phenotypes.[29] Further studies are warranted to elucidate the distinct pathophysiological pathways and clinical implications of nocturnal hypertension and riser BP patterns in this high-risk population.

In the ABP-International study (Ambulatory Blood Pressure in Referred Hypertensive Subjects: an International Database Study), extreme nocturnal dipping was associated with a significantly increased risk of cardiovascular events in individuals aged ≥70 years, particularly among those with low average nighttime SBP (<120 mm Hg), suggesting a U-shaped relationship between nocturnal BP decline and adverse outcomes in the elderly.[33] In contrast, in our study, most patients were receiving antihypertensive treatment, and the association between extreme dipping and cardiovascular risk was attenuated. These findings suggest that overtreatment leading to excessive nocturnal BP reduction may contribute to increased risk in the very elderly, and that

Hypertension FIRST PROOF ONLY

Downloaded from http://ahajournals.org by on February 11, 2026

the presence or absence of antihypertensive treatment might modify this relationship. Careful consideration of treatment intensity and individual nocturnal BP profiles may therefore be especially important in this population.

Implication for Clinical Practice

Our findings underscore the importance of identifying nocturnal hypertension, including its isolated phenotype, as a significant risk factor of composite cardiovascular outcomes in individuals aged ≥80 years. Since nocturnal hypertension frequently occurs in the absence of elevated daytime or office BP,[34] it is often undetected in routine clinical practice. These results support the use of 24-hour ABPM in patients of very advanced age to uncover high-risk phenotypes such as isolated nocturnal hypertension and a riser BP pattern.

In particular, elderly individuals may be more prone to elevated nocturnal BP due to age-related factors such as increased susceptibility to heart failure,[35] reduced renal function,[36] and enhanced salt sensitivity compared with younger populations.[37] Additional contributors in this population include obstructive sleep apnea,[38] autonomic dysfunction such as orthostatic BP fluctuation,[39] a history of cerebrovascular disease,[40] sleep quality,[41] seasonal variations,[42] depression,[43] and cognitive decline,[44] all of which may impair nocturnal BP regulation. Given that older adults are particularly vulnerable to nocturnal hypertension, it is essential to actively identify this high-risk phenotype in routine clinical practice using 24-hour ABPM.

In patients of very advanced age (≥80 years), who are often already being treated with multiple antihypertensive agents or who have increased frailty, strategies to lower nocturnal BP should be implemented cautiously and tailored individually. Particular attention should be paid to the risk of overtreatment and other potential adverse events.[45]

In our analyses, nocturnal DBP demonstrated a U-shaped relationship with composite cardiovascular outcomes in the restricted cubic spline models, but this association was not confirmed when DBP levels were categorized into tertiles. This discrepancy may reflect limited statistical power to detect nonlinear associations when DBP is analyzed in categories, as well as regression dilution due to the relatively narrow distribution of nocturnal DBP values in this cohort.[46] These findings suggest that the prognostic significance of nocturnal DBP is less robust than that of nocturnal SBP, and that further studies are needed to clarify the clinical relevance of nocturnal DBP in elderly populations.

Study Strengths and Limitations

Strengths of this study include its focus on patients of very advanced age (≥80 years), which is typically underrepresented in cardiovascular research, and its real-world, practice-based design. Furthermore, we directly compared the prognostic significance of nocturnal versus daytime hypertension within the same cohort. However, there are also some study limitations. First, this was an observational study, and thus we were unable to determine causality in the findings. Second, 89.2% of our recruited patients were taking antihypertensive medication, and we were unable to assess the association between outcomes and the changes in BP phenotype and the use of antihypertensive medication during follow-up. Third, it is possible that the results of this study should not be extrapolated to individuals with increased frailty, such as inpatient and institutionalized elderly populations. Fourth, ABPM was performed at a single time point; thus, the reproducibility of nocturnal hypertension in patients of very advanced age remains unclear. Fifth, we did not assess sleep quality or sleep-related disorders, such as obstructive sleep apnea or nocturnal urination, either of which may disturb sleep and contribute to nocturnal BP fluctuation.[47,48] Other factors that could not be fully adjusted in this study could potentially have diluted any true association between nocturnal hypertension and the study outcomes. Sixth, the possibility of selection bias cannot be ruled out, as patients who underwent ABPM may have been healthier or more health-conscious than the general very elderly population. Finally, our findings may not be generalized to other racial and ethnic groups.

Conclusions and Perspectives

Nocturnal hypertension was associated with an increased risk of composite cardiovascular outcomes in patients of very advanced age (≥80 years), whereas daytime hypertension was not. The risk of composite cardiovascular outcomes in patients with isolated nocturnal hypertension was comparable to that in those with sustained nocturnal hypertension. These findings highlight the clinical importance of evaluating nocturnal BP in this population. Screening for nocturnal hypertension helps identify a high-risk group for composite cardiovascular outcomes among very old individuals.

Hypertension FIRST PROOF ONLY

Downloaded from http://ahajournals.org by on February 11, 2026

ARTICLE INFORMATION

Received April 29, 2025; accepted December 16, 2025.

Affiliation

Division of Cardiovascular Medicine, Department of Medicine, Jichi Medical University School of Medicine, Shimotsuke, Japan.

Acknowledgments

The authors gratefully acknowledge Kimiyo Saito for coordinating this study and managing the data, and Noriko Harada for her editorial assistance.

Sources of Funding

K. Kario is supported by the 21st Century Center of Excellence Project run by Japan's Ministry of Education, Culture, Sports, Science, and Technology; a grant from the Foundation for Development of the Community (Tochigi, Japan); a grant from

Omron Healthcare, Co, Ltd; a Grant-in-Aid for Scientific Research (B; 21390247) from the Ministry of Education, Culture, Sports, Science and Technology (MEXT) of Japan, 2009 to 2013; and funds from the MEXT-Supported Program for the Strategic Research Foundation at Private Universities, 2011 to 2015 Cooperative Basic and Clinical Research on Circadian Medicine (S1101022). The funding sponsors had no role in designing or conducting this study; in the collection, management, analysis, or interpretation of the data; in the preparation of the article; or in the decision to submit the article for publication.

Disclosures

K. Kario has received research grants from Omron Healthcare, A&D Co. Ltd., and Fukuda Denshi Co. Ltd. The other authors report no conflicts.

Supplemental Material

Tables S1–S7
Figures S1–S2

REFERENCES

1. Blood Pressure Lowering Treatment Trialists' Collaboration. Blood pressure-lowering treatment based on cardiovascular risk: a meta-analysis of individual patient data. *Lancet*. 2014;384:591–598. doi: 10.1016/S0140-6736(14)61212-5
2. Ettehad D, Emdin CA, Kiran A, Anderson SG, Callender T, Emberson J, Chalmer J, Rodgers A, Rahimi K. Blood pressure lowering for prevention of cardiovascular disease and death: a systematic review and meta-analysis. *Lancet*. 2016;387:957–967. doi: 10.1016/S0140-6736(15)01225-8
3. Whelton PK, Carey RM, Aronow WS, Casey DE Jr, Collins KJ, Dennison Himmelfarb C, DePalma SM, Gidding S, Jamerson KA, Jones DW, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the prevention, detection, evaluation, and management of high blood pressure in adults: executive summary: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Hypertension*. 2018;71:1269–1324. doi: 10.1161/HYP.0000000000000066
4. Umemura S, Arima H, Arima S, Asayama K, Dohi Y, Hirooka Y, Horio T, Hoshide S, Ikeda S, Ishimitsu T, et al. The Japanese Society of Hypertension Guidelines for the Management of Hypertension (JSH 2019). *Hypertens Res*. 2019;42:1235–1481. doi: 10.1038/s41440-019-0284-9
5. Mancia G, Kreutz R, Brunström M, Burnier M, Grassi G, Januszewicz A, Muiesan ML, Tsioufis K, Agabiti-Rosei E, Algharably EAE, et al. 2023 ESH Guidelines for the management of arterial hypertension The Task Force for the Management of Arterial Hypertension of the European Society of Hypertension: Endorsed by the International Society of Hypertension (ISH) and the European Renal Association (ERA). *J Hypertens*. 2023;41:1874–2071. doi: 10.1097/HJH.0000000000003480
6. National Institute for Health and Care Excellence. Hypertension in adults: diagnosis and management. NICE guideline. NG136. Accessed April 2025. https://www.nice.org.uk/guidance/ng136
7. Kario K, Hoshide S, Mizuno H, Kabutoya T, Nishizawa M, Yoshida T, Abe H, Katsuya T, Fujita Y, Okazaki O, et al; JAMP Study Group. Nighttime blood pressure phenotype and cardiovascular prognosis: practitioner-based nationwide JAMP Study. *Circulation*. 2020;142:1810–1820. doi: 10.1161/CIRCULATIONAHA.120.049730
8. Staplin N, de la Sierra A, Ruilope LM, Emberson JR, Vinyoles E, Gorostidi M, Ruiz-Hurtado G, Segura J, Baigent C, Williams B. Relationship between clinic and ambulatory blood pressure and mortality: an observational cohort study in 59 124 patients. *Lancet*. 2023;401:2041–2050. doi: 10.1016/S0140-6736(23)00733-X
9. Clement DL, De Buyzere M, De Bacquer DA, de Leeuw PW, Duprez DA, Fagard RH, Gheeraert PJ, Missault LH, Braun JJ, Six RO, et al; The Office versus Ambulatory Pressure Study Investigators. Prognostic value of ambulatory blood-pressure recordings in patients with treated hypertension. *N Engl J Med*. 2003;348:2407–2415. doi: 10.1056/NEJMoa022273
10. Beckett NS, Peters R, Fletcher AE, Staessen JA, Liu L, Dumitrascu D, Stoyanovsky V, Antikainen RL, Nikitin Y, Anderson C, et al; HYVET Study Group. Treatment of hypertension in patients 80 years of age or older. *N Engl J Med*. 2008;358:1887–1898. doi: 10.1056/NEJMoa0801369
11. Williamson JD, Supiano MA, Applegate WB, Berlowitz DR, Campbell RC, Chertow GM, Fine LJ, Haley WE, Hawfield AT, Ix JH, et al; SPRINT Research Group. Intensive vs standard blood pressure control and cardiovascular disease outcomes in adults aged ≥75 years: a randomized clinical trial. *JAMA*. 2016;315:2673–2682. doi: 10.1001/jama.2016.7050
12. Zhang W, Zhang S, Deng Y, Wu S, Ren J, Sun G, Yang J, Jiang Y, Xu X, Wang TD, et al; STEP Study Group. Trial of intensive blood-pressure control in older patients with hypertension. *N Engl J Med*. 2021;385:1268–1279. doi: 10.1056/NEJMoa2111437
13. Benetos A, Labat C, Rossignol P, Fay R, Rolland Y, Valbusa F, Salvi P, Zamboni M, Manckoundia P, Hanon O, et al. Treatment with multiple blood pressure medications, achieved blood pressure, and mortality in older nursing home residents: the PARTAGE study. *JAMA Intern Med*. 2015;175:989–995. doi: 10.1001/jamainternmed.2014.8012
14. Blood Pressure Lowering Treatment Trialists' Collaboration. Age-stratified and blood-pressure-stratified effects of blood-pressure-lowering pharmacotherapy for the prevention of cardiovascular disease and death: an individual participant-level data meta-analysis. *Lancet*. 2021;398:1053–1064. doi: 10.1016/S0140-6736(21)01921-8
15. Fujiwara T, Hoshide S, Kanegae H, Eguchi K, Kario K. Exaggerated blood pressure variability is associated with memory impairment in very elderly patients. *J Clin Hypertens (Greenwich)*. 2018;20:637–644. doi: 10.1111/jch.13231
16. Japanese Society of Hypertension. Japanese Society of Hypertension guideline for the management of hypertension (JSH2004). *Hypertens Res*. 2006;29(Suppl):S1–S105. doi: 10.1291/hypres.29.s1
17. O'Brien E, Parati G, Stergiou G, Asmar R, Beilin L, Bilo G, Clement D, de la Sierra A, de Leeuw P, Dolan E, et al; European Society of Hypertension Working Group on Blood Pressure Monitoring. European Society of Hypertension position paper on ambulatory blood pressure monitoring. *J Hypertens*. 2013;31:1731–1768. doi: 10.1097/HJH.0b013e328363e964
18. KDIGO. Chapter 1: definition and classification of CKD. *Kidney Int Suppl (2011)*. 2013;3:19–62. doi: 10.1038/kisup.2012.64
19. Fujiwara T, Hoshide S, Kanegae H, Kario K. Prognostic value of a riser pattern of nighttime blood pressure in very elderly adults of ≥80 years: a general practice-based prospective SEARCH study. *Am J Hypertens*. 2020;33:520–527. doi: 10.1093/ajh/hpz197
20. de la Sierra A, Redon J, Banegas JR, Segura J, Parati G, Gorostidi M, de la Cruz JJ, Sobrino J, Llisterri JL, Alonso J, et al; Spanish Society of Hypertension Ambulatory Blood Pressure Monitoring Registry Investigators. Prevalence and factors associated with circadian blood pressure patterns in hypertensive patients. *Hypertension*. 2009;53:466–472. doi: 10.1161/HYPERTENSIONAHA.108.124008
21. Kario K, Matsuo T, Kobayashi H, Imiya M, Matsuo M, Shimada K. Nocturnal fall of blood pressure and silent cerebrovascular damage in elderly hypertensive patients: advanced silent cerebrovascular damage in extreme dippers. *Hypertension*. 1996;27:130–135. doi: 10.1161/01.hyp.27.1.130
22. Chobanian AV, Bakris G, Black HR, Cushman WC, Green LA, Izzo JL Jr, Jones DW, Materson BJ, Oparil S, Wright JT Jr, et al; National Heart, Lung, and Blood Institute Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure; National High Blood Pressure Education Program Coordinating Committee. The Seventh Report of the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure: the JNC 7 report. *JAMA*. 2003;289:2560–2572. doi: 10.1001/jama.289.19.2560
23. D'Agostino RB Sr, Vasan RS, Pencina MJ, Wolf PA, Cobain M, Massaro JM, Kannel WB. General cardiovascular risk profile for use in primary care: the Framingham Heart Study. *Circulation*. 2008;117:743–753. doi: 10.1161/CIRCULATIONAHA.107.699579
24. Roush GC, Fagard RH, Salles GF, Pierdomenico SD, Reboldi G, Verdecchia P, Eguchi K, Kario K, Hoshide S, Polonia J, et al; The ABC-H Investigators. Prognostic impact from clinic, daytime, and night-time systolic blood pressure in nine cohorts of 13,844 patients with hypertension. *J Hypertens*. 2014;32:2332–2340, discussion 2340. doi: 10.1097/HJH.0000000000000355
25. Fan HQ, Li Y, Thijs L, Hansen TW, Boggia J, Kikuya M, Björklund-Bodegård K, Richart T, Ohkubo T, Jeppesen J, et al; International Database on Ambulatory Blood Pressure In Relation to Cardiovascular Outcomes Investigators. Prognostic value of isolated nocturnal hypertension on ambulatory measurement in 8711 individuals from 10 populations. *J Hypertens*. 2010;28:2036–2045. doi: 10.1097/HJH.0b013e32833b49fe
26. Wang C, Li Y, Zhang J, Ye Z, Zhang Q, Ma X, Peng H, Lou T. Prognostic effect of isolated nocturnal hypertension in Chinese patients with nondialysis chronic kidney disease. *J Am Heart Assoc*. 2016;5:e004198. doi: 10.1161/JAHA.116.004198
27. de la Sierra A, Staplin N, Ruilope LM, Gorostidi M, Vinyoles E, Segura J, Baigent C, Williams B. A blunted nocturnal blood pressure decline is associated with all-cause and cardiovascular mortality. *J Hypertens*. 2024;42:1197–1202. doi: 10.1097/HJH.0000000000003712

Downloaded from http://ahajournals.org by on February 11, 2026

Hypertension FIRST PROOF ONLY

28. Fukuda M, Munemura M, Usami T, Nakao N, Takeuchi O, Kamiya Y, Yoshida A, Kimura G. Nocturnal blood pressure is elevated with natriuresis and proteinuria as renal function deteriorates in nephropathy. *Kidney Int*. 2004;65:621–625. doi: 10.1111/j.1523-1755.2004.00419.x
29. Huart J, Persu A, Lengele JP, Krzesinski JM, Jouret F, Stergiou GS. Pathophysiology of the nondipping blood pressure pattern. *Hypertension*. 2023;80:719–729. doi: 10.1161/HYPERTENSIONAHA.122.19996
30. Okada Y, Galbreath MM, Shibata S, Jarvis SS, VanGundy TB, Meier RL, Vongpatanasin W, Levine BD, Fu Q. Relationship between sympathetic baroreflex sensitivity and arterial stiffness in elderly men and women. *Hypertension*. 2012;59:98–104. doi: 10.1161/HYPERTENSIONAHA.111.176560
31. Oliveros E, Patel H, Kyung S, Fugar S, Goldberg A, Madan N, Williams KA. Hypertension in older adults: assessment, management, and challenges. *Clin Cardiol*. 2020;43:99–107. doi: 10.1002/clc.23303
32. Frame AA, Wainford RD. Mechanisms of altered renal sodium handling in age-related hypertension. *Am J Physiol Renal Physiol*. 2018;315:F1–F6. doi: 10.1152/ajprenal.00594.2017
33. Palatini P, Verdecchia P, Beilin LJ, Eguchi K, Imai Y, Kario K, Ohkubo T, Pierdomenico SD, Saladini F, Schwartz JE, et al. Association of extreme nocturnal dipping with cardiovascular events strongly depends on age. *Hypertension*. 2020;75:324–330. doi: 10.1161/HYPERTENSIONAHA.119.14085
34. Li Y, Wang JG. Isolated nocturnal hypertension: a disease masked in the dark. *Hypertension*. 2013;61:278–283. doi: 10.1161/HYPERTENSIONAHA.111.00217
35. Savarese G, Lund LH. Global public health burden of heart failure. *Card Fail Rev*. 2017;3:7–11. doi: 10.15420/cfr.2016:25:2
36. Zhang Y, Yu C, Li X. Kidney aging and chronic kidney disease. *Int J Mol Sci*. 2024;25:6585. doi: 10.3390/ijms25126585
37. Nishimoto M, Griffin KA, Wynne BM, Fujita T. Salt-Sensitive hypertension and the kidney. *Hypertension*. 2024;81:1206–1217. doi: 10.1161/HYPERTENSIONAHA.123.21369
38. Li Y, Wang Y. Obstructive sleep apnea-hypopnea syndrome as a novel potential risk for aging. *Aging Dis*. 2021;12:586–596. doi: 10.14336/AD.2020.0723
39. Freeman R, Abuzinadah AR, Gibbons C, Jones P, Miglis MG, Sinn DI. Orthostatic hypotension: JACC State-of-the-art review. *J Am Coll Cardiol*. 2018;72:1294–1309. doi: 10.1016/j.jacc.2018.05.079
40. Liberale L, Badimon L, Montecucco F, Luscher TF, Libby P, Camici GG. Inflammation, aging, and cardiovascular disease: JACC review topic of the week. *J Am Coll Cardiol*. 2022;79:837–847. doi: 10.1016/j.jacc.2021.12.017
41. Oume M, Obayashi K, Asai Y, Ogura M, Takeuchi K, Tai Y, Kurumatani N, Saeki K. Objective sleep quality and night-time blood pressure in the general elderly population: a cross-sectional study of the HEIJO-KYO cohort. *J Hypertens*. 2018;36:601–607. doi: 10.1097/HJH.0000000000001569
42. Kimura T, Senda S, Masugata H, Yamagami A, Okuyama H, Kohno T, Hirao T, Fukunaga M, Okada H, Goda F. Seasonal blood pressure variation and its relationship to environmental temperature in healthy elderly Japanese studied by home measurements. *Clin Exp Hypertens*. 2010;32:8–12. doi: 10.3109/10641960902929479
43. Lorenzo EC, Kuchel GA, Kuo CL, Moffitt TE, Diniz BS. Major depression and the biological hallmarks of aging. *Ageing Res Rev*. 2023;83:101805. doi: 10.1016/j.arr.2022.101805
44. Arvanitakis Z, Shah RC, Bennett DA. Diagnosis and management of dementia: Review. *JAMA*. 2019;322:1589–1599. doi: 10.1001/jama.2019.4782
45. Sheppard JP, Koshiaris C, Stevens R, Lay-Flurrie S, Banerjee A, Bellows BK, Clegg A, Hobbs FDR, Payne RA, Swain S, et al. The association between antihypertensive treatment and serious adverse events by age and frailty: A cohort study. *PLoS Med*. 2023;20:e1004223. doi: 10.1371/journal.pmed.1004223
46. Moore MN, Atkins ER, Salam A, Callisaya ML, Hare JL, Marwick TH, Nelson MR, Wright L, Sharman JE, Rodgers A. Regression to the mean of repeated ambulatory blood pressure monitoring in five studies. *J Hypertens*. 2019;37:24–29. doi: 10.1097/HJH.0000000000001977
47. Xu J, Ding N, Zhang X, Wang N, Sun B, Zhang R, Xie X, Wan Z, Gu Y, Zhang S, et al. Nocturnal blood pressure fluctuation and associated influential factors in severe obstructive sleep apnea patients with hypertension. *Sleep Breath*. 2018;22:1045–1052. doi: 10.1007/s11325-018-1634-6
48. McKeigue PM, Reynard JM. Relation of nocturnal polyuria of the elderly to essential hypertension. *Lancet*. 2000;355:486–488. doi: 10.1016/S0140-6736(00)82039-5

Hypertension
FIRST PROOF ONLY

Downloaded from http://ahajournals.org by on February 11, 2026